Question 1: (a) Using Binet's formula, we compute that

$$f_1 = \frac{\alpha-\beta}{\alpha-\beta} = \frac{\sqrt{5}}{\sqrt{5}} = 1, \text{ and } f_2 = \frac{\alpha^2-\beta^2}{\alpha-\beta} = \frac{\frac{1+\sqrt{5}}{2}+1-\frac{1-\sqrt{5}}{2}-1}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{5}} = 1.$$

(c) *Proof.* Assume $\alpha = \frac{1+\sqrt{5}}{2}$, $\beta = \frac{1-\sqrt{5}}{2}$, and $f_n = \frac{\alpha^n-\beta^n}{\alpha-\beta}$. Consider $f_{n+1} = f_n + f_{n-1}$. This can rewritten as

$$f_n + f_{n-1} = \frac{\alpha^n-\beta^n}{\alpha-\beta} + \frac{\alpha^{n-1}-\beta^{n-1}}{\alpha-\beta} = \frac{\alpha^{n-1}(\alpha+1)-\beta^{n-1}(\beta+1)}{\alpha-\beta}.$$

We know that $\alpha+1 = \alpha^2$ and $\beta+1 = \alpha^2$, so we can rewrite the expression as

$$\frac{\alpha^{n-1}(\alpha+1)-\beta^{n-1}(\beta+1)}{\alpha-\beta} = \frac{\alpha^{n-1}\alpha^2-\beta^{n-1}\beta^2}{\alpha-\beta} = \frac{\alpha^{n+1}-\beta^{n+1}}{\alpha-\beta}.$$

Thus, we are done. □

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Question 2: For the following proofs, let R be a commutative ring and $x, y, z \in R$.

(a) *Proof.* We show that $(-x)(-y)+(-(xy)) = 0$, and thus $(-x)(-y) = xy$. As was shown in Proposition 5.7 in the notes, $-(xy) = (-x)y$. So, we compute that

$$(-x)(-y)+(-(xy)) = (-x)(-y)+(-x)y = (-x)(-y+y) = (-x)\cdot 0 = 0.$$

Thus, we are done. □

(b) *Proof.* Considering $x(y-z) = x(y+(-z))$, we can appeal to the distributive property of commutative rings. So, $x(y+(-z)) = xy + x(-z)$. Appealing to Proposition 5.7 in the notes, we have that $x(-z) = -(xz)$, we get that $xy + x(-z) = xy - xz$. □

(c) *Proof.* Considering $-1(x)$, we can appeal to Proposition 5.7 in the notes, and rewrite the expression as $-(1x)$. Appealing to R's multiplicative identity, which is 1, we know $1x = x, \forall x \in R$. Thus, $-(1x) = -x$. □

Question 3: *Proof.* Suppose R is an ordered ring, $x, y, z \in R, x < y, 0 < z$. It follows that $y-x \in R^+$ and $z \in R^+$. Appealing to the closure of positives, $(y-x)(z) \in R^+$. Distributing, we get that $yz - xz \in R^+$. Thus, $xz < yz$. □

Question 4: *Proof.* For the sake of contradiction, assume that there is a largest element of $\mathbb{Z}$ called a such that $a \in \mathbb{Z}^+$. Let $x \in \mathbb{Z}^+$. Consider $x + a$; by closure of the positives, $x + a \in \mathbb{Z}^+$. Consider $(x+a) - a = x + a + (-a) = x \in \mathbb{Z}^+$. Thus, $a + x > a$, which is a contradiction because we assumed a was the largest element of $\mathbb{Z}$. Thus, there is no largest element of $\mathbb{Z}$. Similarly, the smallest element of $\mathbb{Z}$ is the additive inverse of the largest element of $\mathbb{Z}$, which we have showed to not exist, thus there is no smallest element of $\mathbb{Z}$.

□

Question 5: *Proof.* We proceed by induction.

BC When $n = 0$, we have that $\sum_{k=0}^{0} k^2 = 0 = \frac{n(n+1)(2n+1)}{6}\Big|_{n=0}$. So, the base case is true.

IS Assume $\sum_{k=0}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$. Consider $\sum_{k=0}^{n+1} k^2$. We can rewrite this as

$$\sum_{k=0}^{n+1} k^2 = (\sum_{k=0}^{n} k^2) + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2.$$

Rearranging, we find that this is equivalent to $\frac{2n^3+9n^2+13n+6}{6}$. Thus,

$$(\sum_{k=0}^{n} k^2) + (n+1)^2 = \frac{2n^3+9n^2+13n+6}{6}.$$

We find that this is further equivalent to $\frac{(n+1)(n+2)(2(n+1))}{6}$. Thus,

Thus, $\sum_{k=0}^{0} k^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$. ☐

Question 6: *Proof.* We proceed by induction.

BC When $n = 1$, we have that $\sum_{k=0}^{1} r^k = r + 1 = \frac{r^{(1+1)}-1}{r-1}$.

IS Assume that $\sum_{k=0}^{n} r^k = \frac{r^{n+1}-1}{r-1}$. Consider $\sum_{k=0}^{n+1} r^k$. This can be rewritten as $(\sum_{k=0}^{n} r^k) + r^{n+1}$. We know that $\sum_{k=0}^{n} r^k = \frac{r^{n+1}-1}{r-1}$, we can rewrite as

$$\frac{r^{n+1}-1}{r-1} + r^{n+1} = \frac{r^{n+1}-1+r^{n+2}-r^{n+1}}{r-1} = \frac{r^{n+2}-1}{r-1}.$$

Thus, we are done.

☐

Question 7: We claim that $\sum_{k=1}^{n}(2k-1) = n^2, \forall n \in \mathbb{Z}$.

Proof. Summation is a linear operator, so $\sum_{k=1}^{n}(2k-1) = 2\sum_{k=1}^{n} k - \sum_{k=1}^{n} 1$. The first term is a known result from class: $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. The second term is simply n. So, we get that $\sum_{k=1}^{n}(2k-1) = n(n+1) - n = n^2$. ☐

Question 8: We claim that $\sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{n}{n+1}, \forall n \in \mathbb{Z}$. We provide two different proofs.

Proof. First, we can proceed via algebraic manipulation. We find that $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$. Rewriting, we find that the sum of these terms is a basic telescopic series:

$$\sum_{k=1}^{n}(\tfrac{1}{k} - \tfrac{1}{k+1}) = (\tfrac{1}{1} - \tfrac{1}{2}) + (\tfrac{1}{2} - \tfrac{1}{3}) + (\tfrac{1}{3} - \tfrac{1}{4}) \dots (\tfrac{1}{n} - \tfrac{1}{n+1}).$$

We see that every term cancels with its opposite, other than 1 and $\frac{1}{n+1}$. Thus, $\sum_{k=1}^{n}(\frac{1}{k} - \frac{1}{k+1}) = 1 - \frac{1}{n+1} = \frac{n}{n+1}$. □

Proof. Another method of proof is by induction.

BC When $n = 1$, we have that $\sum_{k=1}^{1} = \frac{1}{2} = \frac{n}{n+1}\Big|_{n=1}$.

IS Assume that $\sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{n}{n+1}$. Consider $\sum_{k=1}^{n+1} \frac{1}{k(k+1)}$. This can be rewritten:

$$\sum_{k=1}^{n+1} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \frac{1}{k(k+1)} + \frac{1}{(n+1)(n+2)} = \frac{(n+1)^2}{(n+1)(n+2)}.$$

Simplifying, we see that $\frac{(n+1)^2}{(n+1)(n+2)} = \frac{n+1}{n+2}$. So, $\sum_{k=1}^{n+1} \frac{1}{k(k+1)} = \frac{n+1}{n+2}$.

Thus, we are done. □

Question 9: *Proof.* We proceed by induction.

BC When $n = 7$, $7! = 5040 > 2187 = 3^7$. Thus, the base case is true.

IS Assume that $n! > 3^n$. Then, consider $(n+1)!$ and 3^{n+1}. These can be rewritten as $(n+1)n!$ and $3(3^n)$, respectively.

Lemma 1. *If $x > y > 0$ and $z > w > 0$, then $xz > yw$.*

Proof. We showed in Question 3 that if $x > y$ and $z > 0$, then $xz > yz$. Consider $yz - yw = y(z - w)$. By assumption we have $z > w$, so $z - w \in \mathbb{Z}^+$. Also by assumption, $y \in \mathbb{Z}$. Appealing to the closure of the positives, $y(z-w) \in \mathbb{Z}^+$. Thus, by definition, $yz > yw$. So, by transitivity, $xz > yz > yw \implies xz > yw$. □

We can see that $n+1 > 3$ when $n \geq 7$, and we have that $n! > 3^n$ for $n \geq 7$ as an assumption. Thus, by Lemma 1, we have that $(n+1)n! > 3(3^n)$ for $n \geq 7$. Thus, $(n+1)! > 3^{n+1}$ for $n \geq 7$, so we are done.

□